MILITARY SPECIFICATION

NETWORK C: 10526476

1. SCOPE

1.1 This specification covers one type of network circuit board consisting of the pulse stretcher, 10 cycle oscillator, neon driver, teletype oscillator, strobe circuit, and a battery anode supply circuit. The network is used as part of the Gun Direction Computer: M18.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18 Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component parts

DRAWINGS

U.S. Army Munitions Command

F10526476	Assembly, Computer Subassembly - Network C

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PACKAGING DATA SHEET

M14002 Data Sheet, Master Packaging

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication.- The network shall be manufactured in accordance with Drawing F10526476 and drawings pertaining thereto.

3.2 General specifications.- The network shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.2.1 MIL-F-13926

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Workmanship

3.2.2 MIL-C-46357

(a) Etched circuit assembly
(b) Marking
(c) Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The network shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the network for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the network shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The network shall withstand 3 shock impulses in the vertical plane relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3 Temperatures.

3.3.3.1 Operating.- The network shall meet the requirements of 3.4.1.4, 3.4.4.3, 3.4.5.2, 3.4.6.2, and 3.4.7.5 while exposed to temperatures of -40°F and +135°F.

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3.3.3.2 Storage.- The network shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

3.3.4 Humidity.

3.3.4.1 Operating.- The network shall meet the requirements of 3.4.1.4, 3.4.4.3, 3.4.5.2, 3.4.6.2 and 3.4.7.5 while exposed to an atmosphere with a relative humidity (RH) up to 97%, at a temperature of +135°F.

3.3.4.2 Non-operating.- The network shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The network shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The network shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a fine spray of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Performance.- Unless otherwise specified the network shall meet the performance requirements of this specification at a standard ambient temperature between 60° and 90°F and at a barometric pressure of 28 to 32 inches of mercury with a RH up to 90%.

3.4.1 Neon driver No. 1.

3.4.1.1 Input diodes.- The input diodes shall have a forward voltage drop of less than 0.8 volt and a reverse leakage of less than 50 microamperes.

3.4.1.2 True state loaded.- The base to emitter forward voltage drop of the input transistor shall be less than 1.1 volts.

3.4.1.3 False state loaded.- The voltage measured from pin 17B to ground shall be 112 ± 15 Vdc under the following conditions:

(a) With a load (Figure 1) connected to pin 17B.
(b) With pin 4A grounded.
(c) With the appropriate voltages connected to the pins designated below:

TO PIN	VOLTAGE SUPPLY
48A	-1.5V
8B	-3.0V
24A	-6.0V
30A	-18V
61B	+1.25V

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TO PIN	VOLTAGE SUPPLY
29A	+3.0V
33A	+6.0V
20B	+15V
2B	+35V
14A	+155V
12A	+66V
19B	+143V
39B	GND
40B	GND
48B	-10V

3.4.1.4 Dynamic output.- The output waveform shall exhibit the characteristics shown in Figure 2 when supplied with an input waveform as shown in Figure 2. The delay time, measured from the time the input is at minus 3 volts (going towards minus 6 volts), to the time the output reaches plus 90 volts (going towards plus 75 volts) shall be less than 10 milliseconds. The delay time measured from the time the input is at minus 3 volts (going towards zero volts), to the time the output reaches plus 90 volts (going towards plus 110 volts), shall be less than 10 milliseconds.

3.4.2 BAD supply and zener voltage.- The zener diode breakdown voltage shall be 6.3 $\pm$ 0.5 volts.

3.4.3 Resistor value.- The resistance value of these resistors shall not vary more than $\pm$ 1 percent from their rated values.

3.4.4 Pulse stretcher.

3.4.4.1 False state loaded.- When the input to the pulse stretcher is false, the following conditions shall exist:

(a) The input switch transistor collector leakage shall be less than 40 microamperes.

(b) The ground clamp diode at the base of the trigger transistor shall have a forward voltage drop not exceeding 0.5 volt.

(c) The trigger transistor emitter-to-base leakage shall be less than 25 microamperes.

(d) The base-to-emitter forward voltage drop of the stable switch transistor shall not exceed 0.5 volt.

(e) The forward voltage drop of CR27 and CR28 shall not exceed 0.7 volt.

(f) The saturation voltage of the stable switch transistor shall be less than 0.4 volt.

(g) The collector leakage current of the unstable switch transistor shall be less than 40 microamperes.

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(h) The forward voltage drop of the CR32 and CR33 shall be less than 0.8 volt.

(i) The collector leakage current of the driver transistor shall be less than 25 microamperes.

3.4.4.2 True state loaded.- When the input to the pulse stretcher is true, the following conditions shall exist:

(a) The saturation and the base-to-emitter voltage drop of the input switch transistor shall be less than 0.50 and 0.65 volt respectively.

(b) The forward voltage drop of CR29 and CR30 shall be less than 0.75 volt.

(c) The saturation and the base-to-emitter voltage drop of the unstable switch transistor shall be less than 0.45 and 0.65 volt, respectively.

(d) The saturation and the base-to-emitter voltage drop of the inverter transistor shall be less than 0.45 and 0.65 volt, respectively.

(e) The saturation and the base-to-emitter voltage drop of the driver transistor shall be less than 0.55 and 0.65 volt, respectively.

3.4.4.3 Dynamic output.- The output waveform shall exhibit the characteristics shown in Figure 3, when supplied with an input test signal as shown in Figure 3. The delay time measured from the time the input is at minus 1.5 volts (going towards minus 6 volt), to the time the output reaches minus 1.5 volts (going towards minus 3.5 volts), shall be less than 2 microseconds. The output true time pulse width measured at the minus 2 volts level shall be 10 $\pm$ 2.5 microseconds.

3.4.5 10 CPS oscillator.

3.4.5.1 Static test.- When the 10 cps oscillator is tested statically, the following conditions shall exist:

(a) The transistor collector leakage currents shall be less than 25 microamperes.

(b) The transistor saturation voltages shall be less than 0.30 volt.

(c) The base-to-emitter voltages shall be less than 0.60 volt.

(d) The diode forward voltage drops shall be less than 0.90 volt.

3.4.5.2 Dynamic output.- When the 10 cps oscillator is loaded and appropriate power is applied, the following conditions exist:

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(a) The output from the astable multivibrator shall be a waveform that goes to 0 ± 0.5 volt from minus 9.6 ± 1 volts and remains there for 55 ± 25 milliseconds before dropping to minus 5.5 ± 1 volts. The voltage then returns to minus 10 ± 1 volts from minus 5.5 ± 1.0 volts in 55 ± 25 milliseconds. (See Figure 4B).

(b) The output of the 10 cps oscillator shall be a square wave that alternates between 0 ± 1 volt and minus 18 ± 2 volts. The period shall be 110 ± 50 milliseconds and each half period shall be 55 ± 25 milliseconds. (See Figure 4A).

3.4.6 Strobe.

3.4.6.1 Static test.

(a) The saturation voltage of transistor Q2 shall be less than 0.3 volt dc.

(b) The voltage from the collector of Q2 to the collector of Q1 shall be less than 0.8 volt dc when both transistors are on.

(c) The forward voltage drop across CR11 shall be less than 0.7 volt dc.

(d) When transistor Q2 is off, the strobe output shall be in the -9 ± 1.0 volts dc state.

(e) When transistor Q1 is off, the strobe output shall be in the -9.5 ± 1.0 volts dc state.

3.4.6.2 Dynamic test.- When the specified input signal is applied to the strobe circuit, the output will be as specified in Figure 5.

3.4.7 Teletype oscillator.

3.4.7.1 Padding resistance.- Resistors R1, R3 and R8 must be selected in functional tests. These three resistors determine the frequency of the output square wave.

3.4.7.2 Dynamic conditions for unijunction transistor

(a) The standoff ratio of the unijunction transistor shall be between 0.48 and 0.65.

(b) The saturation voltage of the emitter to base one junction shall be less than 4.0 volts.

3.4.7.3 Inverter circuit.

(a) The input diode shall clip the pulse to 2 volts.

(b) The forward voltage drop of the base clamping diode shall be between +0.1 volt and +0.5 volt.

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(c) The saturation voltage shall be less than 0.3 volts.

(d) The forward voltage drop of the clamp diode shall be less than 1 volt.

(e) The output pulse shall meet the requirements of Figure 6.

3.4.7.4 Flip-Flop circuit.

(a) The saturation voltage of both transistors shall be less than 0.3 volts.

(b) The base-to-emitter voltage of the "off" transistor shall be 0.7 $\pm$ 0.4 volt.

(c) The forward voltage drop of the clamp diodes shall be less than 1 volt.

(d) The test diodes shall have a leakage of less than 30 micro-amperes at 6 volts reverse voltage and a forward voltage drop of less than 1 volt.

(e) The output waveform under full load shall fulfill the requirements of Figure 7.

3.4.7.5 Dynamic output.- The cirucit output, when energized from the appropriate dc power supplies, shall exhibit the characteristics of the waveform shown in Figure 7.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

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4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables I, II and III and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance & Rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirement and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE I

(100 percent inspection)

Characteristic	Requirement	Test Procedures
101. Input diodes (Neon driver No. 1)	3.4.1.1	4.5.2.1.1
102. True state loaded (Neon driver No. 1)	3.4.1.2	4.5.2.1.2
103. False state loaded (Neon driver No. 1)	3.4.1.3	4.5.2.1.3
104. Dynamic output (Neon driver No. 1)	3.4.1.4	4.5.2.1.4
105. Bad supply and zener voltage	3.4.2	4.5.2.2
106. Resistor value	3.4.3	4.5.2.3
107. False state loaded (Pulse stretcher)	3.4.4.1	4.5.2.4.1
108. True state loaded (Pulse stretcher)	3.4.4.2	4.5.2.4.2
109. Dynamic output (Pulse stretcher)	3.4.4.3	4.5.2.4.3
110. Static test (10 cps Oscillator)	3.4.5.1	4.5.2.5.1
111. Dynamic output (10 cps Oscillator)	3.4.5.2	4.5.2.5.2
112. Static test (Strobe)	3.4.6.1	4.5.2.6.1
113. Dynamic test (Strobe)	3.4.6.2	4.5.2.6.2
114. Dynamic conditions for unijunction transistors (Teletype oscillator)	3.4.7.2	4.5.2.7.1
115. Padding resistance (Teletype oscillator)	3.4.7.1	4.5.2.7.2
116. Flip-flop circuit (Teletype oscillator)	3.4.7.4	4.5.2.7.3
117. Inverter circuit (Teletype oscillator)	3.4.7.3	4.5.2.7.4
118. Dynamic output (Teletype oscillator)	3.4.7.5	4.5.2.7.5

4.3 Control tests.

4.3.1 General.- One network shall be selected at random as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table II.

TABLE II

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301. Fabrication	3.1	Applicable drawings - visual
302. General specifications	3.2	MIL-F-13926 and MIL-C-46357 - visual

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4.3.2 Environmental.- Three units shall be selected at random as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

TABLE III

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
303. Vibration	3.3.1	4.5.1.1
304. Shock	3.3.2	4.5.1.2
305. Humidity-Operating	3.3,4.1, 3.4.1.4, 3.4.4.3, 3.4.5.2, 3.4.6.2, 3.4.7.5	4.5.1.3.2, 4.5.2.1.5, 4.5.2.4.3.4, 4.5.2.5.2.3, 4.5.2.6.3, 4.5.2.7.5.2
306. Humidity-Non-operating	3.3.4.2	4.5.1.3.1
307. Salt spray	3.3.6	4.5.1.4
308. Operating temperature	3.3.3.1, 3.4.1.4, 3.4.4.3, 3.4.5.2, 3.4.6.2, 3.4.7.5	4.5.1.5, 4.5.2.1.5, 4.5.2.4.3.4, 4.5.2.5.2.3, 4.5.2.6.3, 4.5.2.7.5.2
309. Storage temperature	3.3.3.2	4.5.1.5

4.3.3 Control test failure.- Should any one item of a control test sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

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Item Number	Description
1	Oscilloscope - A dual trace oscilloscope with a rise time of less than 50 millimicroseconds and an input impedance of 8 megohms or greater and an input capacitance of less than 25 micromicrofarad. Sweep speed of 50 millimicrosecond per centimeter and greater.
2	Voltage measuring device - A ± 1 percent voltage measuring device with a 10 millivolt to 300 volt range. The input impedance shall be greater than 10 megohms.
3	Pulse generator - A generator of pulses capable of square waves up to 100 kc with a rise and fall time of less than 1 microsecond and an output impedance of less than 150 ohms.
4	Power supply - Minus 1.5 volts ± 2 percent with a current requirement of minus 150 milliamperes. The peak-to-peak ripple and noise shall be less than 100 millivolts. The output impedance shall be less than 2 ohms from dc to 500 kc.
5	Power supply - Minus 3 volts ± 2 percent current requirement of minus 50 milliamperes. The peak-to-peak ripple and noise shall be less than 100 millivolts. The output impedance shall be less than 2 ohms from dc to 500 kc.
6	Power supply - Minus 6 volts ± 2 percent, current requirement of 200 milliamperes. The peak-to-peak ripple and noise shall be less than 50 millivolts and the output impedance shall be less than 2 ohms from dc to 500 kc.
7	Power supply - Minus 18 volts ± 2 percent, current requirement of 150 milliamperes. The peak-to-peak ripple and noise shall be less than 600 millivolts and the output impedance shall be less than 2 ohms from dc to 500 kc.
8	Power supply - Plus 3 volts ± 2 percent, current requirement of 50 milliamperes. The peak-to-peak ripple and noise shall be less than 100 millivolts and the output impedance shall be less than 2 ohms from dc to 500 kc.
9	Power supply - Plus 6 volts ± 2 percent, current requirement of 150 milliamperes. The peak-to-peak ripple and noise shall be less than 100 millivolts and the output impedance shall be less than 2 ohms from dc to 500 kc.
10	Power supply - Plus 15 volts ± 2 percent, current requirement of 50 milliamperes. The peak-to-peak ripple and noise shall be less than 100 millivolts and the output impedance shall be less than 2 ohms from dc to 500 kc.

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Item Number	Description
11	Power supply - Plus 155 volts ± 2 percent, current requirement of 120 milliamperes. The peak-to-peak ripple and noise shall be less than 400 millivolts and the output impedance shall be less than 10 ohms from dc to 500 kc.
12	Power supply - Plus 66 volts minus 12, plus 9 percent, current requirement of 120 milliamperes. The peak-to-peak ripple and noise shall be less than 200 millivolts and the output impedance shall be less than 10 ohms from dc to 500 kc.
13	Power supply - Variable, from zero to one hundred and fifty volts current requirement of 100 milliamperes. The peak-to-peak ripple and noise shall be less than 500 millivolts and the output impedance shall be less than 10 ohms from dc to 500 kc.
14	Power supply - Plus 143 volts shall not vary more than 12.5 ± 1 volts from the plus 155 volts (Item 11) supply. The current requirement of 50 milliamperes and a peak-to-peak ripple and noise of less than 400 millivolts. The output impedance shall be less than 10 ohms from dc to 500 kc.
15	Power supply - Plus 1.25 volts ± 2 percent, current requirement of 100 milliamperes. The peak-to-peak ripple and noise shall be less than 100 millivolts and the output impedance shall be less than 2 ohms from dc to 500 kc.
16	Power supply - Plus 35 volts ± 1 percent and a current requirement of 50 milliamperes. The peak-to-peak ripple and noise shall be less than 100 millivolts. The output impedance shall be less than 2 ohms from dc to 500 kc.
17	Frequency counter - Shall be capable of determing the period of the 11.110 millisecond square wave to six significant figures, with an accuracy of plus or minus one in the last significant figure. The input capacitance to the counter shall be less than 75 micromicrofarad and the resistance shall be greater than 100 k ohms.
18	Current measuring device - Capable of determining 0 to 100 microamperes with a % 5 percent full scale accuracy. The total input impedance shall be less than 500 ohms.
19	Pulse generator - Pulse repetition rate of 448,000 ± 10 percent pulses per second. The pulse base line shall be plus 0.2 ± 0.2 volt. The pulse amplitude shall be plus 5.0 ± 0.5 volts. The 10 to 90 percent rise and fall time of the pulse shall be less than 0.3 microseconds. The 50 percent amplitude duration of the pulse shall be 0.5 ± 0.1 microsecond.

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Item Number	Description
20	Power supply - minus 10 volts + 2 percent, current requirement of 150 milliamperes. The peak-to-peak ripple and noise shall be less than 100 millivolts and the output impedance shall be less than 2 ohms from dc to 500 KC.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the network rigidly mounted in its normal operating position. Subsequent to the vibration test, the network shall show no damage or defect resulting from the test when examined by visual and tactile means. The network shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock.- This test shall be conducted in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the network shall be shocked in the vertical plane only. The impact (shocks) and "G" value shall be in accordance with 3.3.2. A receiving device, representing the connector receptacle that receives the network in the computer, shall be secured to the horizontal surface of the shock tester. Each end of the receiving device shall be clamped such that the female receiving slot will hold the unit in a vertical plane. The unit will then be inserted into the receiving device and shocked. Subsequent to imposing the number of impacts required, a visual examination shall be made for damages or defects to determine compliance with 3.3.2 and then the unit will be inspected as specified in 4.5.1.3 prior to determining compliance with 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 97 $\pm$ 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with paragraph 4.5.2 of this specification, and shall meet the specified requirements of 3.3.4.1.

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4.5.1.4 Salt spray.- The network shall be subjected to the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the logic driver shall be inspected for any physical damage, then tested in accordance with 4.5.2 for compliance with 3.3.6.

4.5.1.5 Temperature (Operating and storage).- After being subjected to all environmental tests specified above, the network shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the network shall be tested in accordance with 4.5.2.1.4, 4.5.2.4.3, 4.5.2.5.2, 4.5.2.6.3, 4.5.2.6.2, 4.5.2.7.5.2 to determine compliance with the requirements as specified in 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the network shall be tested in accordance with 4.5.2 to 4.5.2.7.5 inclusive to determine compliance with 3.3.3.2.

4.5.1.6 Fungus.- One of the first 5 networks produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Performance.- Unless otherwise specified, the network shall meet the performance tests of this specification at the conditions specified in 3.4.

4.5.2.1 Neon driver No. 1.

4.5.2.1.1 Input diodes test.- The following tests shall be conducted to determine the forward voltage drop and reverse leakage as specified in 3.4.1.1.

4.5.2.1.1.1 Forward voltage drop.- Connect the minus 18 volt supply to pin 30A. Ground pins 4A, 6A, 10B, and 15B one at a time for each test and the voltage measured at pin 7B shall be minus 0.45 $\pm$ 0.35 volt dc.

4.5.2.1.1.2 Reverse leakage.- Remove the minus 18 volts from pin 30A and ground pin 30A. The voltage measured at pin 7B, when the minus 18 volts is applied in sequence to pins 4A, 6A, 10B and 15B, shall be more positive than minus 0.106 volts dc. Remove all test leads after completion of tests.

4.5.2.1.2 True state loaded.- The base to emitter forward voltage drop shall meet the requirement of 3.4.1.2 when tested as follows.

4.5.2.1.2.1 Load.- Connect to pin 17B the load shown in Figure 1.

4.5.2.1.2.2 Power supplies.- Connect power supplies to the appropriate pins as specified in Table IV.

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TABLE IV

To Pin	Power Supply	Item No.
48A	-1.5v	4
8B	-3.0v	5
24A	-6.0v	6
30A	-18v	7
61B	+1.25v	15
29A	+3.0v	8
33A	+6.0v	9
20B	+15v	10
2B	+35v	16
14A	+155v	11
12A	+66v	12
19B	+143v	14
39B, 40B	Ground	
48B	-10v	21

4.5.2.1.2.3 True input.- Apply the minus 6 volt supply to pin 4A. The voltage measured from the pins listed in column 1 to those in column 2 shall be as specified in column 3 of Table V. Remove the minus 6 volt supply from pin 4A.

TABLE V

1 From Pin	2 To Pin	3 Voltage
8B	7B	+0.6 ± 0.5
39B	11B	-23 ± 15
39B	17B	-75 ± 9

4.5.2.1.3 False state loaded.- The following tests shall be conducted to voltage measurements for determining the requirements of 3.4.1.3.

4.5.2.1.3.1 False input.- With the load and power supplies connected (see 4.5.2.1.2.1 and 4.5.2.1.2.2), ground pin 4A. The voltage measured from ground to pin 17B, shall be plus 112 ± 15 volts dc. Remove the ground connection at pin 4A.

4.5.2.1.4 Dynamic output.- Dynamic output for the neon driver No. 1 shall be tested for compliance with 3.4.1.4 by connecting the load and power supplies (see 4.5.2.1.2.1 and 4.5.2.1.2.2). Apply a square wave of 10 cps ±10 percent varying from 0 ± 1 volt to minus 6 ± 2 volts as shown in Figure 2 to pin 4A. The output waveform at pin 17B, shall be as shown in Figure 2. The delay time as shown in Figure 2, measured from the time the input is at minus 3 volts (going towards minus 6 volts), to the time the output reaches plus 90 volts, shall be less than 10 milliseconds. The delay time measured from the time the input is at minus 3 volts (going towards zero volts), to the time the output reaches plus 90 volts (going towards plus 110 volts), shall be less than 10 milliseconds. At the end of these tests remove all test leads and power supply connections.

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4.5.2.1.5 Circuit output.- The neon driver No. 1 shall be tested in accordance with 4.5.2.1.4 to determine compliance with 3.4.1.4 as required by 3.3.3.1 and 3.3.4.1 when thermally stabilized at ambient temperature of minus 40 degrees to plus 135 degrees Fahrenheit. At the end of these tests remove all test leads and power supplies.

4.5.2.2 BAD supply and zener voltage.- To determine compliance with 3.4.2, ground pin 39B and apply to pin 19B the plus 143 volt supply Voltage. The voltage measured from pin 19B to pin 14B shall be plus 6.3 ± 0.5 volts dc. Remove all power supplies.

4.5.2.3 Resistor value.- Resistor value variance as specified in 3.4.3 shall be determined by connecting to each pin 50B, 46B, and 46A, a 2.8k ohm 1/2 watt, ± 1 percent resistor from ground. Apply minus 10 volts to pin 48B. The voltage measured at those pins in column 1 shall be as specified in column 2 of Table VI. At the end of these tests remove all test leads and power supplies.

TABLE VI

1 Pin	2 Voltage
50B	-2.9 ± 0.3
46B	-5.0 ± 0.5
46A	-6.1 ± 0.6

4.5.2.4 Pulse stretcher.

4.5.2.4.1 False state loaded.- The following tests shall be conducted to determine compliance with 3.4.4.1.

4.5.2.4.1.1 Load.- Connect a 150 ohm, 1 watt, ± 5 percent resistor from ground to pin 36A. Connect an identical resistor from ground to pin 50A.

4.5.2.4.1.2 Power supply.- Connect power supplies to appropriate pins, as specified in Table IV.

4.5.2.4.1.3 False input.- Ground pins 40A and 56B. The voltage measured from those points listed in column 1 to those in column 2 shall be as specified in column 3 of Table VII.

TABLE VII

1 From Pin	2 To Pin	3 Voltage
45B	48B	Less positive than +0.4
49B	48B	Less positive than +0.4
39B	44A	Less positive than +0.5
39B	54B	Less positive than +0.5
TP8	38A	Less negative than -0.012
TP9	55A	Less negative than -0.012
29A	TP8	Less positive than +0.5
29A	TP9	Less positive than +0.5

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TABLE VII (Continued)

1	2	3
From Pin	To Pin	Voltage
29A	TP10	Less positive than +0.4
29A	58A	Less positive than +0.4
TP8	TP4	Less positive than +0.7
TP9	TP6	Less positive than +0.7
37A	30A	Less positive than +0.6
57A	30A	Less positive than +0.6
24A	TP13	Less positive than +0.8
24A	TP14	Less positive than +0.8
36A	35B	Less positive than +0.01
50A	52A	Less positive than +0.01

4.5.2.4.2 True state loaded.- The following tests shall be conducted to determine compliance with 3.4.4.2.

4.5.2.4.2.1 Load and power supply.- The load and power supplies shall remain connected as specified in 4.5.2.4.1.1 and 4.5.2.4.1.2.

4.5.2.4.2.2 True input.- Apply minus 6 volts to pins 40A and 56B. Apply plus 6 volts to pins 38A and 55A. The voltage measured from the points listed in column 1, to those in column 2, shall be as specified in column 3, of Table VIII.

TABLE VIII

1	2	3
From Pin	To Pin	Voltage
48A	43A	Less positive than +0.65
48A	51B	Less positive than +0.65
48A	45B	Less positive than +0.50
48A	49B	Less positive than +0.50
29A	TP8	Less negative than -0.75
29A	TP9	Less negative than -0.75
39B	37A	Less positive than +0.45
39B	57A	Less positive than +0.45
39B	TP11	Less positive than +0.65
39B	TP12	Less positive than +0.65
39B	45A	Less positive than +0.45
39B	49A	Less positive than +0.45
39B	TP3	Less positive than +0.65
39B	TP5	Less positive than +0.65
24A	TP13	Less negative than -0.65
24A	TP14	Less negative than -0.65
24A	35B	Less negative than -0.65
24A	52A	Less negative than -0.65

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4.5.2.4.3 Dynamic output.- Compliance with 3.4.4.3 shall be determined by connecting the load and power supplies (see 4.5.2.4.1.1 and 4.5.2.4.1.2). Apply a 5 kc ± 10 percent, square wave varying from - ± 1 to minus 4 ± 2 volts to pins 40A and 56B, as shown in Figure 3. The 10 to 90 percent rise and fall time shall be less than or equal to 4 microseconds.

4.5.2.4.3.1 Trigger input waveform.- The trigger input waveform at pins 44A and 54B shall be as specified in Figure 3. The delay time measured from the time the input is at minus 1.5 volts (going towards minus 4 volts) to the time the trigger input waveform is at plus 5 volts (going towards plus 6 volts) shall be less than 0.5 microsecond. The pulse width measured at the 10 percent point shall be 8.0 ± 2.0 microseconds.

4.5.2.4.3.2 Stable switch output waveform.- The stable switch output waveform at TP10 and pin 58A shall be as specified in Figure 3. The delay time measured from the time the input is at minus 1.5 volts (going towards minus 4 volts) to the time the stable switch output waveform is at 1.5 volts (going towards minus 3 volts) shall be less than or equal to 1.5 microseconds. The pulse width measured at the zero volts level shall be 8.0 ± 2.0 microseconds.

4.5.2.4.3.3 Driver output waveform.- The driver output waveform at pins 36A and 50A shall be as specified in Figure 3. The delay time measured from the time the input is at minus 1.5 volts (going towards minus 4 volts) to the time the driver output waveform is at minus 1.5 volts (going towards minus 3.5 volts) shall be less than or equal to 2 microseconds. The pulse width measured at the minus 2 volts level shall be 10.0 ± 2.5 microseconds. At the end of these tests remove all test leads and power supply connections.

4.5.2.4.3.4 Circuit output.- The pulse stretcher output waveform shall be tested in accordance with 4.5.2.4.3.3 and Figure 3 at pins 36A and 50A, to determine compliance with 3.4.4.3 as required by 3.3.3.1 and 3.3.4.1 when thermally stabilized at ambient temperatures of minus 40 degrees to plus 135 degrees Fahrenheit. At the end of these tests remove all test leads and power supply connections.

4.5.2.5 10 cps oscillator performance.

4.5.2.5.1 Static tests.- The following tests shall be conducted to determine compliance with 3.4.5.1.

4.5.2.5.1.1 Power supply connections and loading.- Connect the dc power supplies to the indicated pins shown in Table IV. Connect a 1K ohm ± 5 percent, 1/2 watt resistor from minus 18 volts to pin 20A. (See Figure 8).

4.5.2.5.1.2 Q21 leakage test.- Ground pin 23A. Measure minus 5 ± 4 volts at pin 13A.

4.5.2.5.1.3 Diode test.- The voltage measure at pin 25A shall be more negative than plus 0.90 volt.

4.5.2.5.1.4 Q21 saturation voltage test.- Apply minus 3 to pin 25B. The voltage measured at pin 13A shall be more positive than minus 0.30 volt.

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4.5.2.5.1.5 Q21 base emitter voltage.- The voltage measured at pin 17A shall be more positive than minus 0.60 volt. Remove minus 3 volts from pin 25B and remove ground from pin 23A.

4.5.2.5.1.6 Q26 leakage test.- Ground pin 13A. Measure minus 5 ± 4 volts at pin 23A.

4.5.2.5.1.7 Diode test.- Measure a voltage more negative than plus 0.90 volt at pin 17A.

4.5.2.5.1.8 Q26 saturation voltage.- Apply minus 3 volts to test point 7. The voltage measured at pin 23A shall be more positive than minus 0.30 volt.

4.5.2.5.1.9 Q26 base-to-emitter test.- The voltage measured at pin 25A shall be more positive than minus 0.60 volt. Remove minus 3 volts from test point 7 and remove ground from pin 13A.

4.5.2.5.1.10 Q27 saturation and base-to-emitter voltage.- Connect minus 3 volts to pin 23A. The voltage measured at pin 20A shall be more positive than minus 0.30 volt. The voltage measured at test point 15 shall be more positive than minus 0.60 volt. Remove minus 3 volts from pin 23A.

4.5.2.5.2 Dynamic output.- The conditions specified in 3.4.5.2 shall be determined by the following tests.

4.5.2.5.2.1 Power supply connections and loading.- Connect the dc power supplies to the indicated pins shown in Table IV. Connect a 1K ± 5 percent 1/2 watt resistor from minus 18 volts to pin 20A. (See Figure 8)

4.5.2.5.2.2 Multivibrator test.- The characteristics of the voltage waveform at pin 23A shall be as shown in Figure 4B. The characteristics at pin 20A shall be as shown in Figure 4A.

4.5.2.5.2.3 Circuit output.- Under the conditions of 4.5.2.5.2.1 the characteristics of the voltage waveform at pin 20A shall be as shown in Figure 4B, to determine compliance with 3.4.5.2 as required by 3.3.3.1 and 3.3.4.1, when thermally stabilized at ambient temperatures of minus 40 degrees to plus 135 degrees Fahrenheit. At the end of these tests remove all test leads and power supplies.

4.5.2.6 Strobe.

4.5.2.6.1 Static test.- To determine compliance with the strobe requirements of 3.4.6.1, the following procedures and tests shall be conducted:

(a) Connect the power supplies specified in column A of Table IX to the respective pins in column B of Table IX. Connect the ground returns of these power supplies to pins 39B and 40B.

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TABLE IX

A Power Supply	B Pin
-18v	30A
-10v	48B
+1.25v	61B
+3.0v	29A

(b) Connect pin 55B to pin 39B through a 56.0 ohm, ± 5 percent, 1/4 watt resistor.

(c) Measure the voltage from the pin indicated in column A to the respective pin in column B of Table X. The voltage is indicated in the same row in column C of Table X.

TABLE X

A	B	C Volts dc
29A	52B	less than 0.3
52B	60B	less than 0.8
60B	62A	less than 0.7

(d) Connect the plus 6v power supply to pin 61A. Measure the voltage from the pin indicated in column A to the respective pin in column B of Table XI. The voltage is indicated in the same row in column C of Table XI. Remove the plus 6v power supply from pin 61A after the voltage measurements have been taken.

TABLE XI

A	B	C Volts dc
29A	52B	plus 3.0 ± 0.5
52B	60B	minus 9.5 ± 1.0
39B	62A	minus 9.0 ± 1.0

(e) Remove the resistor between pins 55B and 39B. Connect the plus 6 volt power supply to pin 55B. Measure the voltage from pin 60B to pin 39B. The voltage shall be minus 10 ± 1.0 vdc.

(f) Measure the voltage from pin 62A to pin 39B. The voltage shall be minus 9.5 ± 1.0 vdc. Remove the plus 6 volt power supply from pin 55B.

4.5.2.6.2 Dynamic test.- This test shall be conducted to determine compliance with 3.4.6.2 by connecting the power supplies as specified in column A to the respective pins in column B of Table IX. Connect the ground returns of these power supplies to pins 39B and 40B. Connect the signal generator specified in paragraph 4.10.2, item 20, through 470 ohm ± 5 percent,

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1/4 watt resistor to pin 55B. The generator output waveform is specified in Figure 9. Connect the load specified in Figure 10 to pins 62A, and 39B and the minus 3 volt power supply as indicated in Figure 10. Connect the oscilloscope (item 1 of 4.4.1.1) to pin 65A. Connect the ground return of the scope to pin 40B. The signal observed on the oscilloscope is shown in Figure 5. Disconnect the load, signal generator and oscilloscope upon completion of tests.

4.5.2.6.3 Circuit output.- The strobe output waveform shall be as specified in Figure 5 to determine compliance with 3.4.6.2 as required by 3.3.3.1 and 3.3.4.1 when thermally stabilized at ambient temperatures of minus 40 degrees to plus 135 degrees Fahrenheit. Disconnect the load, signal generator, and oscilloscope upon completion of tests.

4.5.2.7 Teletype oscillator.- The following tests shall be conducted to determine compliance with 3.4.7.

4.5.2.7.1 Dynamic conditions for unijunction transistor.- The oscillation frequency of the unijunction transistor circuit is determined by the parallel combination of resistors R1, R3, and R8. These three resistors shall be selected in functional test to set the output period at pin 19A to 22.22 $\pm$ .11 millisecond to determine compliance with 3.4.7.2. The frequency shall be measured with the module temperature of 75 degrees Fahrenheit $\pm$ 10 degrees Fahrenheit and the supply voltage at plus 35 $\pm$ 0.35 volts. After any component in the teletype circuit is changed, the output period at pin 19A must remain at 22.22 $\pm$.11 milliseconds or new padding resistors must be selected by the procedures in 4.5.2.7.2.1, 4.5.2.7.2.2 and 4.5.2.7.2.3.

4.5.2.7.2 Padding resistance.- The following three paragraphs will determine the procedure for selecting the padding resistors R1, R3, R8 in compliance with 3.4.7.1. The frequency measurements shall be made at pin 5B. With a counter that will read six significant figures. The padding resistor values with their corresponding ordnance number are given in Table XII.

TABLE XII

VALUES FOR R1	ORD. NUMBERS
84.5 K ohms	10525862
78.7 K ohms	10525861
71.5 K ohms	10525860
66.5 K ohms	10525859
60.4 K ohms	10525858
90.9 K ohms	10525863

VALUES FOR R3	ORD. NUMBERS
619 K ohms	10525866
768 K ohms	10525867
931 K ohms	10525868
1.18 Megohms	10525869
1.62 Megohms	10525870
2.43 Megohms	10525871
5.11 Megohms	10525872

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TABLE XII

VALUES FOR R8	ORD. NUMBERS
8.45 Megohms	10525873
5.11 Megohms	10525872
16.9 Megohms	10525874

4.5.2.7.2.1 Selection of resistor R1.- Connect the dc power and ground to the module pins shown in Table IV. Place the 90.0K, resistor in the R1 location, and measure the period at pin 5B. Repeat the step for the 84.5K, 78.7K, 71.5K, 66.5K, 60.4K resistors. Select the resistor that produces a period between 11.055 millisecond and 12.156 millisecond. Allow the circuit to operate for three minutes. The period at pin 5B shall still be between 11.055 millisecond and 12.156 millisecond, if not, select the next lower value of resistance. If the period is between 11.110 ± 0.055 millisecond; omit paragraphs 4.5.2.7.2.2 and 4.5.2.7.2.3.

4.5.2.7.2.2 Steps in selection of resistor R3.- Place the 5.11 meg. resistor in the R3 location and measure the period at pin 5B. Repeat this step for the 2.43 meg., 1.62 meg., 1.18 meg., 931K, 768K, and the 619K resistors. Select the resistor value if any, that produces period between 11.055 millisecond and 11.313 millisecond. After selection of resistor R3, if the period is 11.110 ± .055 millisecond, omit paragraph 4.5.2.7.2.3.

4.5.2.7.2.3 Steps in selection of resistor R8.- Place the 16.9 meg., resistor in R8 location and measure the period at pin 5B. Repeat this step for the 8.45 meg. and 5.11 meg. resistors. Select a resistor value that will make the period 11.110 ± .055 millisecond. After three minutes of operation, the output at pin 5B should be rechecked for the correct period of 11.110 ± .055 millisecond.

4.5.2.7.3 Flip-Flop circuit.- Compliance with 3.4.7.4 shall be determined by connecting the dc power and ground to the module as listed in Table IV. No input signal is required for the test.

4.5.2.7.3.1 Diode and transistor tests.- Connect pin 39B to pin 12B and minus 6 volts to pin 15A. The output voltage from pin 19A to pin 39B shall be minus 6.0v ± 1.5 volts. The voltage from pin 3A to pin 39B shall be plus .7 ± .4 volt. The voltage from pin 5A to pin 39B shall be minus 0.15 ± 0.15 volt. The voltage from 19A to 24A shall be less than 1 volt. Remove the inputs from pins 12B and 15A.

4.5.2.7.3.2 Diode and transistor tests.- Apply ground to pin 15A and minus 6 volts to pin 12B. The output at pin 19A shall be minus 0.15 ± 0.15 volt. The voltage from pin 5A to pin 39B shall be minus 6.0v ± 1.5 volts. The voltage from pin 7A to pin 39B shall be plus .7 ± .4 volt. The voltage from pin 5A to 24A shall be less than 1 volt. Remove the inputs from pins 12B and 15A.

4.5.2.7.3.3 Diode forward voltage test.- Disconnect the plus 35 volts from the module. The voltage from pin 5B to pin 24A shall be less than 1 volt. The voltage from pin 2A to pin 39B shall be plus .3 ± .2 volt. Remove ground pin from 15A and the minus 6 volts from pin 12B.

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4.5.2.7.4 Inverter circuit.- Compliance with 3.4.7.3 shall be determined by applying dc power and grounds to the module as listed in Table IV. Apply the load shown in Figure 11 to pin 19A. The following paragraphs and Figure 6 describe the waveforms observed at the test points and module pins.

(a) The output of the inverter transistor Q7 is observed at pin 5B see Figure 6c. The output is a positive pulse from minus 6v ± 1.2 volts to minus 0.15 ± 0.15, with a rise time (minus 6 to 0) 10 percent to 90 percent, of .20 ± .15 microseconds. The 50 percent pulse width shall be 8 ± 5 microseconds, with a repetition rate of 11.110 ± .055 microseconds.

(b) The voltage waveform at pin 1B shall be a sawtooth between plus 2v ± 1 volts and plus 20v ± 3 volts, as in Figure 6A. The fall time (20v to 2v) from 90 percent to 10 percent, shall be less than 20 microseconds.

(c) The waveform at pin 1A see Figure 6B shall be a minus 2.0, minus 1.5, plus 1.0 volts negative pulse with a fall time, 0 to minus 1 volt, of less than 3.5 usec. The pulse repetition rate shall be 11.110 ± .055 milliseconds.

4.5.2.7.5 Dynamic output.- Compliance with 3.4.7.5 shall be determined by conducting the following tests.

4.5.2.7.5.1 Connect the dc power and ground to the module pins shown in Table IV. Apply the load shown in Figure 11 to the output pin 19A. The output waveform, see Figure 7A, shall be a negative square wave between minus 6.0v ± 1.0 volts and 0.15 ± 0.15 volt. The square wave shall have a rise and fall time of less than .4 microsecond, measured at the 10 percent to 90 percent points with a repetition rate of 22.22 ± .11 milliseconds.

4.5.2.7.5.2 To determine compliance with 3.4.7.5 as required by 3.3.3.1 and 3.3.4.1 the output waveform at pin 19A, when loaded with the circuit shown in Figure 11, shall possess the characteristics of the waveform shown in Figure 7B at the temperature of minus 40 and plus 135 degrees Fahrenheit as specified in 3.3.3.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with Master Packaging Data Sheet M14002.

6. NOTES

6.1 Intended use.- The Network C circuit board consists of a pulse stretcher, 10 cycle oscillator, neon driver, teletype oscillator, strobe circuit, and a battery anode supply circuit for the Gun Direction Computer: M18. The intended use of each of these circuits is as follows:

(a) The pulse stretcher is used to delay input signals to the computer by a period of 6.2 microseconds or greater.

(b) The ten cps oscillator operates the neon drivers which turn on and off neon tubes on the control panel to indicate malfunction of the computer.

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(c) The neon driver No. I is used to turn on and off neon indicators on the face of the control panel to indicate malfunctions of the computer.

(d) The teletype oscillator employs a unijunction transistor oscillator in connection with a bistable flip-flop to provide a precision time reference for the computer two wire, five level, serial teletype output.

(e) The strobe circuit provides a signal for driving the strobe switches of the reading system. This is a portion of the process for converting the memory read signals to logical signals.

(f) The bad supply is used to reduce the anode supply voltage for the battery display nixie tubes.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet (see 5.1).
(e) A block of serial numbers to identify each networks.

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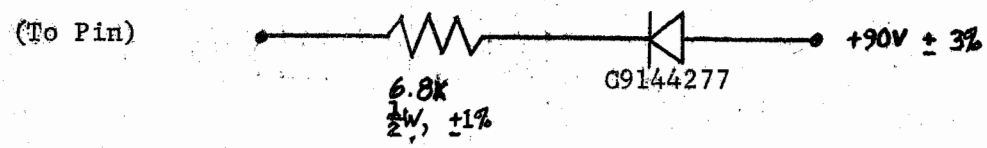
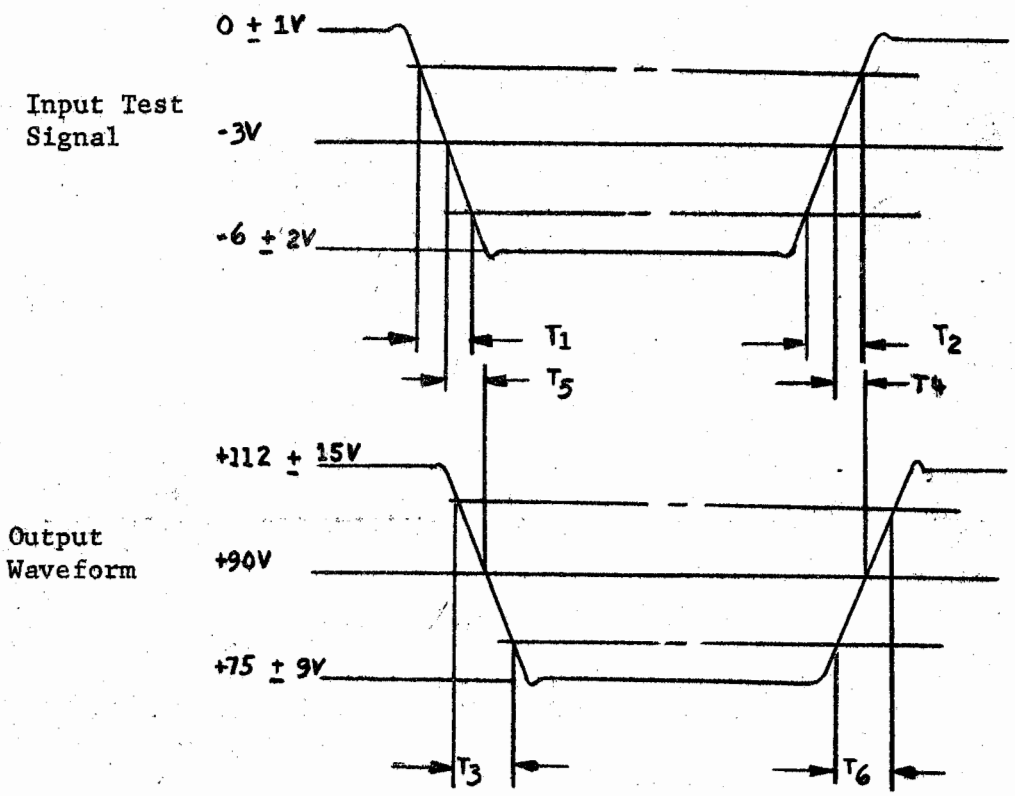


FIGURE I Neon Driver No. 1 LOAD



$T_1 - T_2 \leq 1ms$
$T_3 - T_6 \leq 1ms$
$T_5 - T_4 \leq 10ms$

FIGURE 2 NEON DRIVER NO. I WAVEFORM

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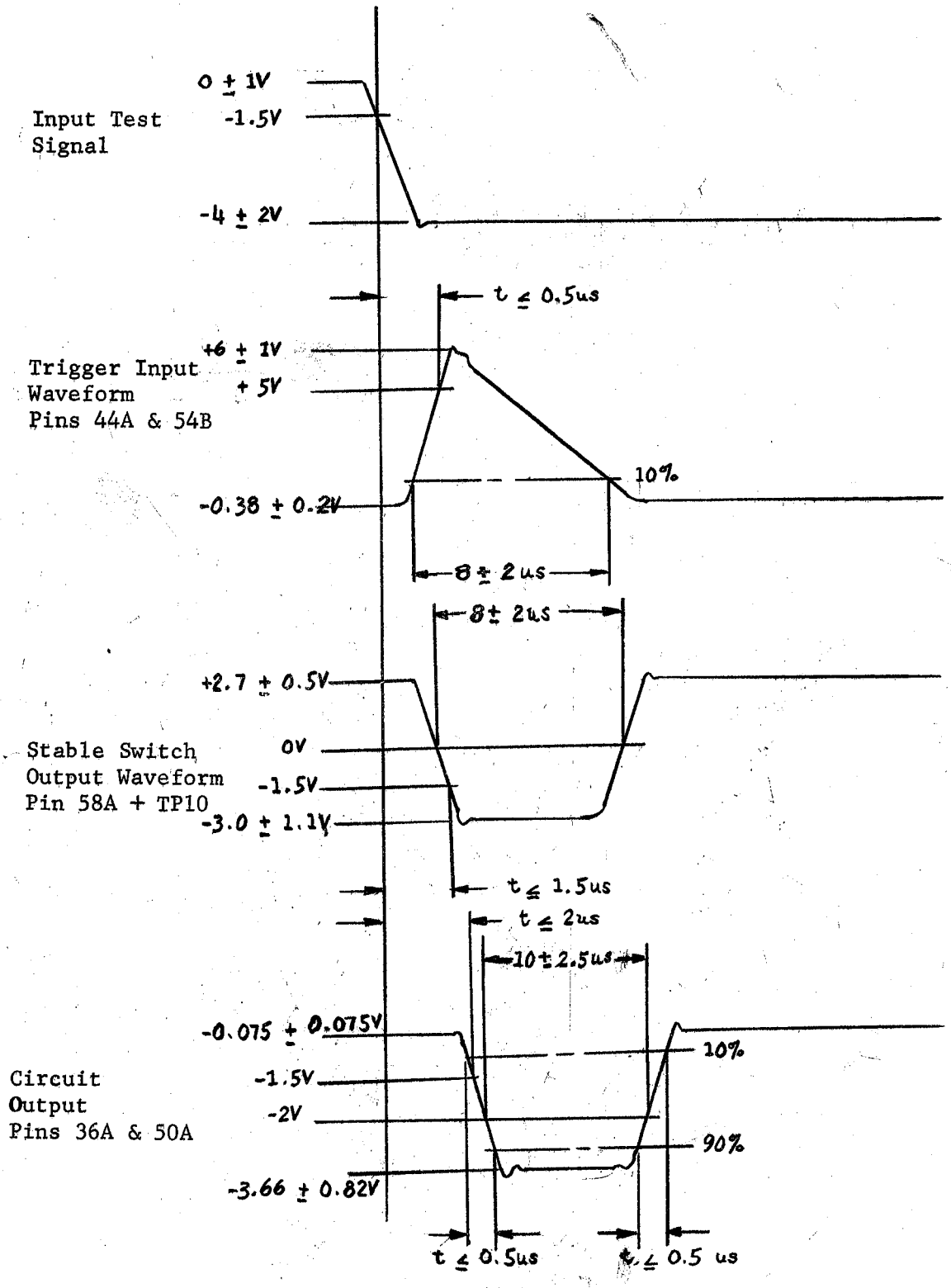


FIGURE 3 PULSE STRETCHER WAVEFORM

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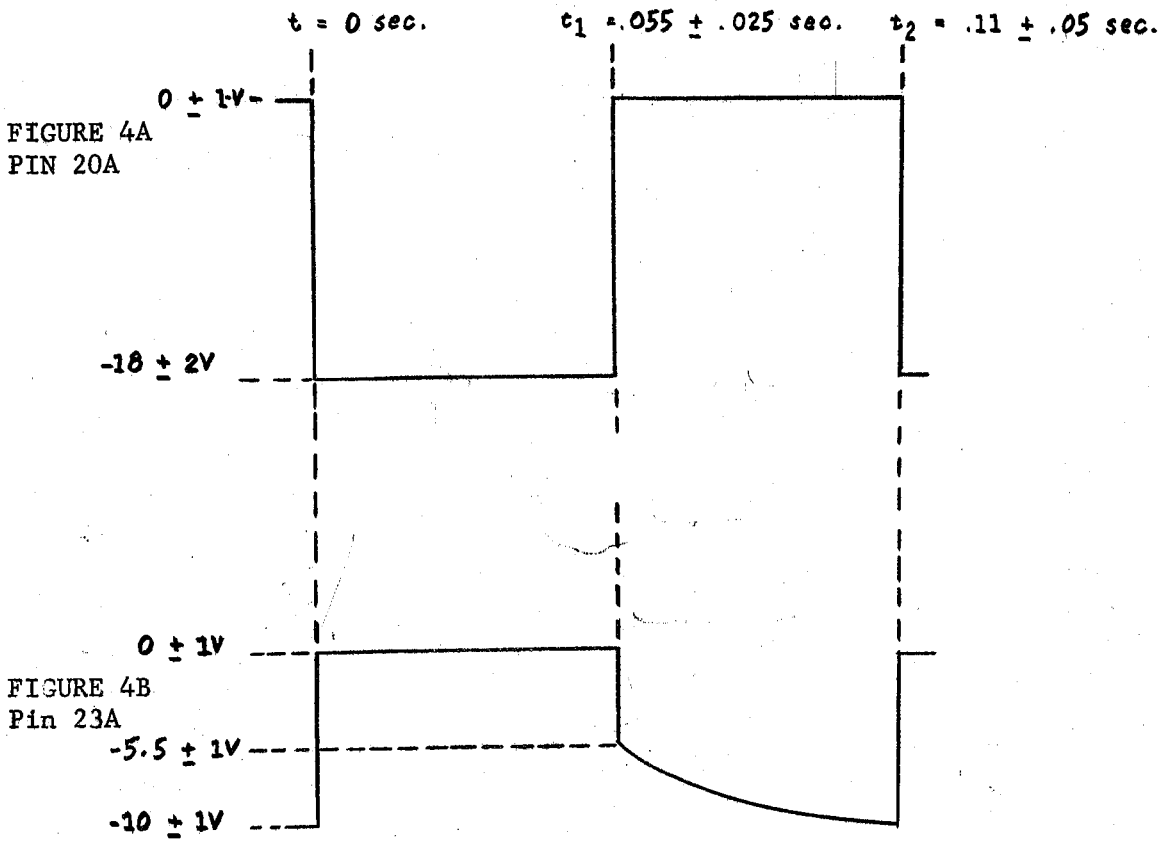


FIGURE 4 10 CPS OSCILLATOR OUTPUT

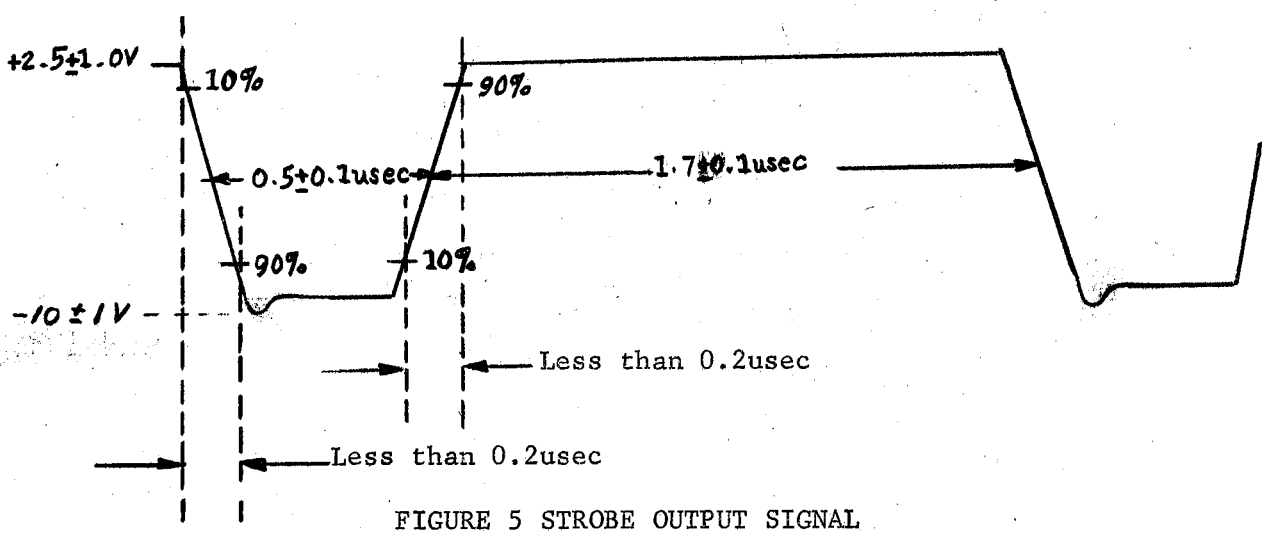


FIGURE 5 STROBE OUTPUT SIGNAL

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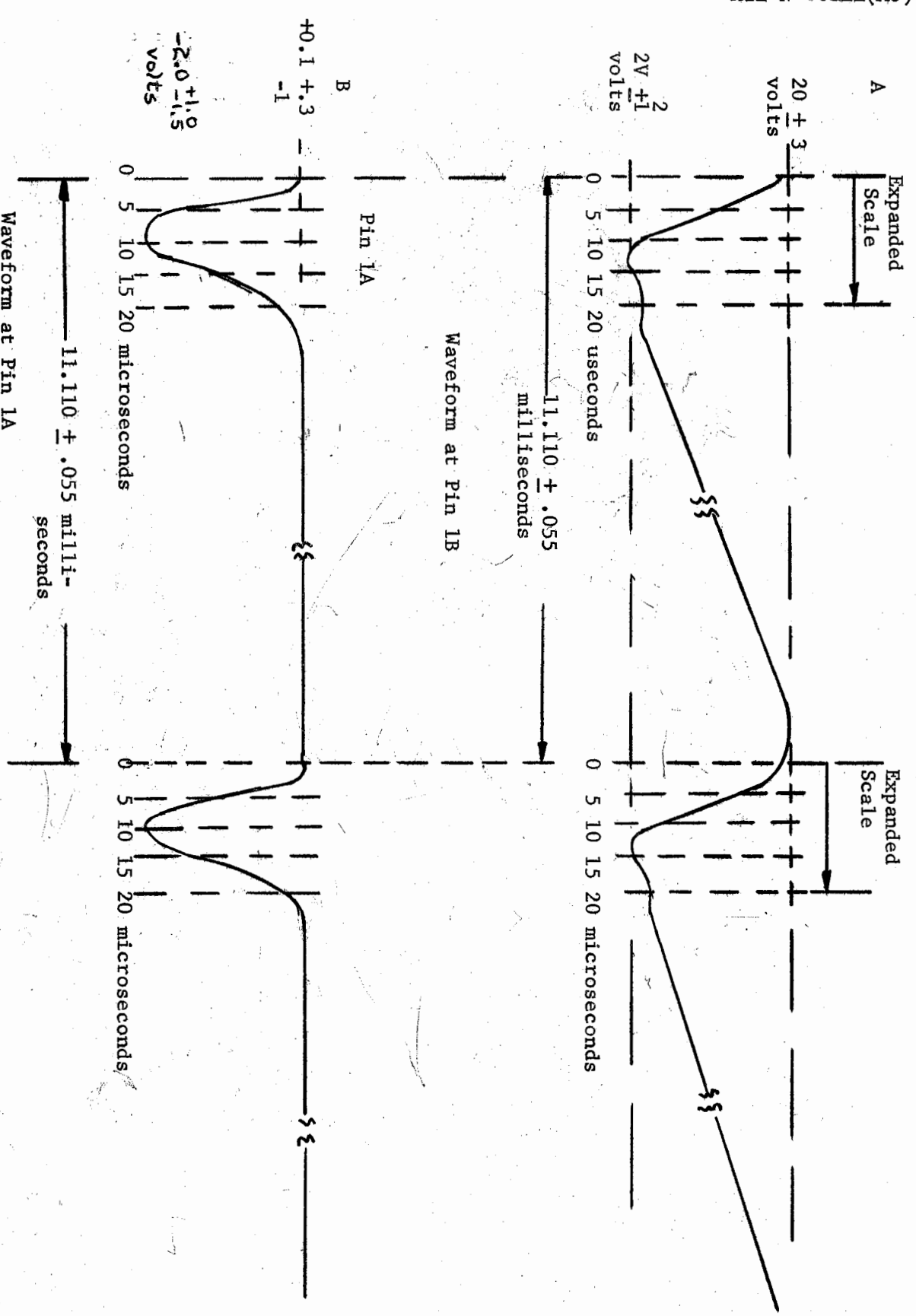


FIGURE 6

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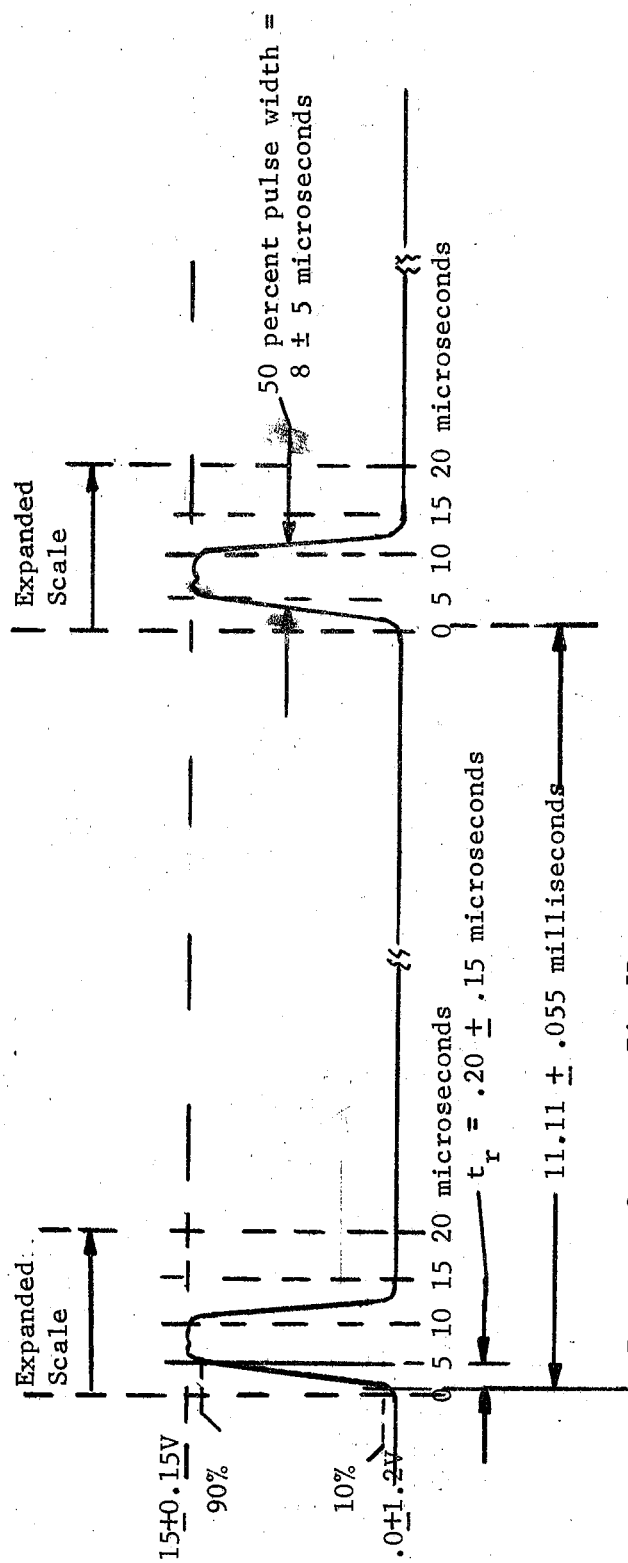
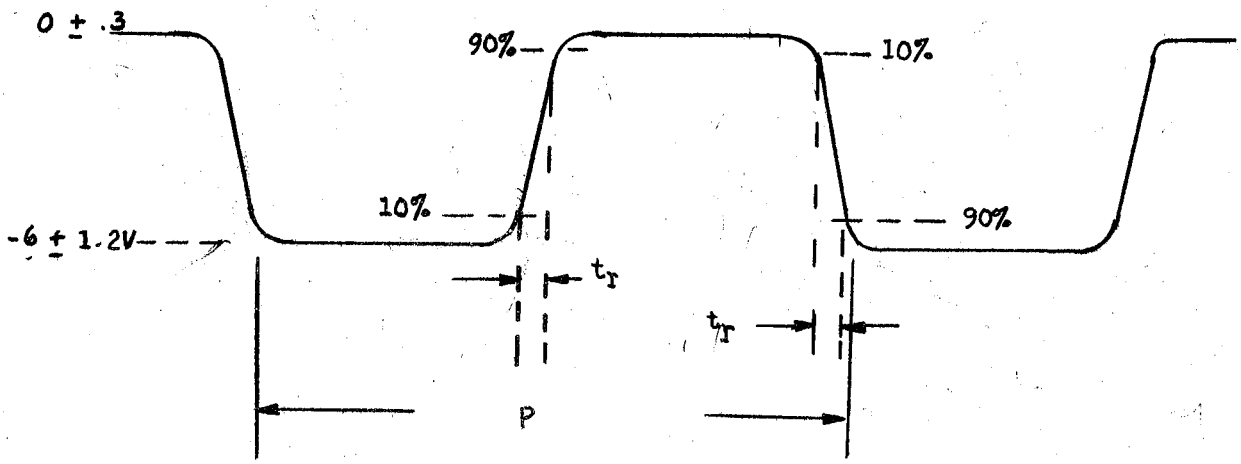


FIGURE 6 Cont'd

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A. $t_r = t_f \leq .4$ microseconds, with Temperature = $75^{\circ}F \pm 10^{\circ}F$
P - 22.22 ± .110 milliseconds $V_{cc} = +35V \pm .35V$

B. $t_r = t_f \leq .6$ microseconds with Temperature from $-40^{\circ}F + 0 + 140^{\circ}F$
P = 22.22 $^{+.880}_{-1.320}$ milliseconds $V_{cc} + 35V \pm 2.5$ volts

FIGURE 7 OUTPUT WAVEFORM

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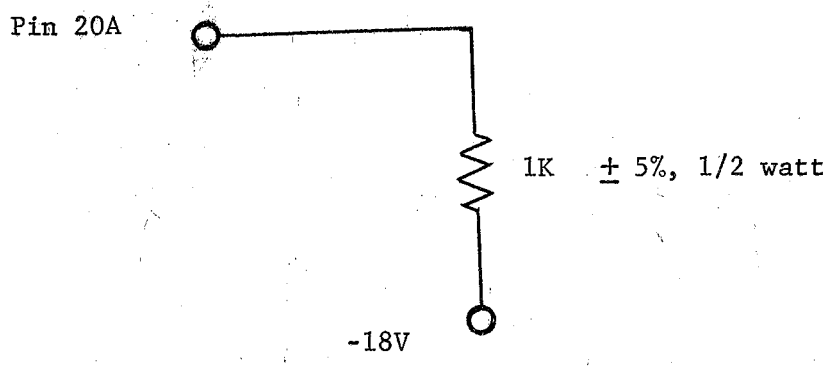


FIGURE 8 LOAD

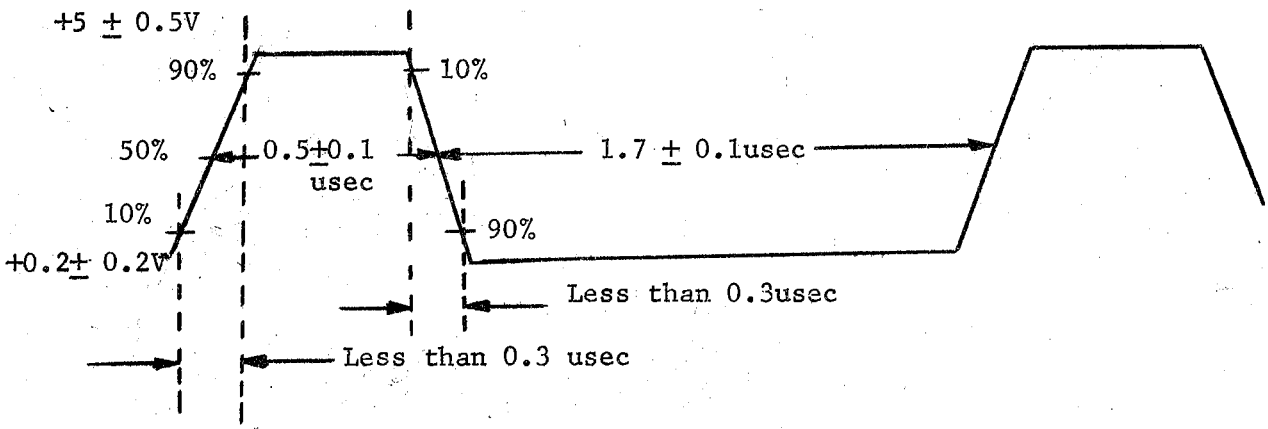


FIGURE 9

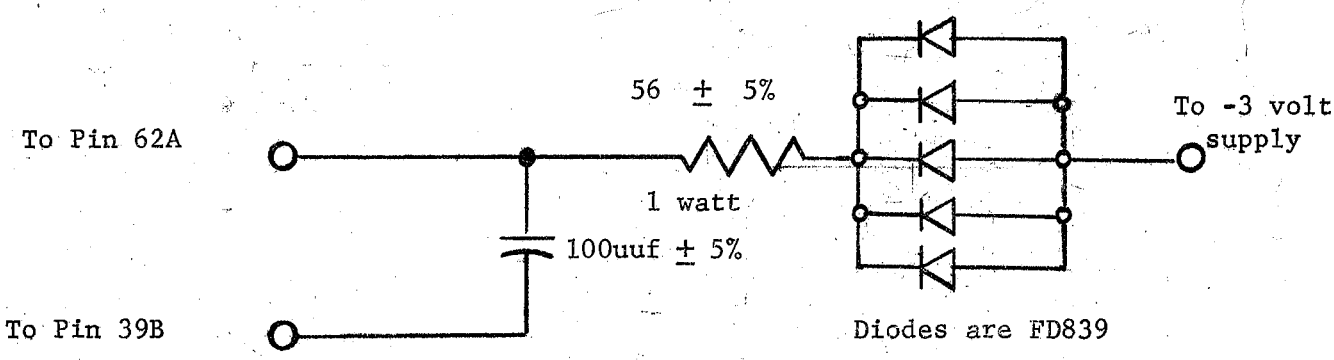


FIGURE 10

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:18Z
Check the source to verify that this is the current version before use.

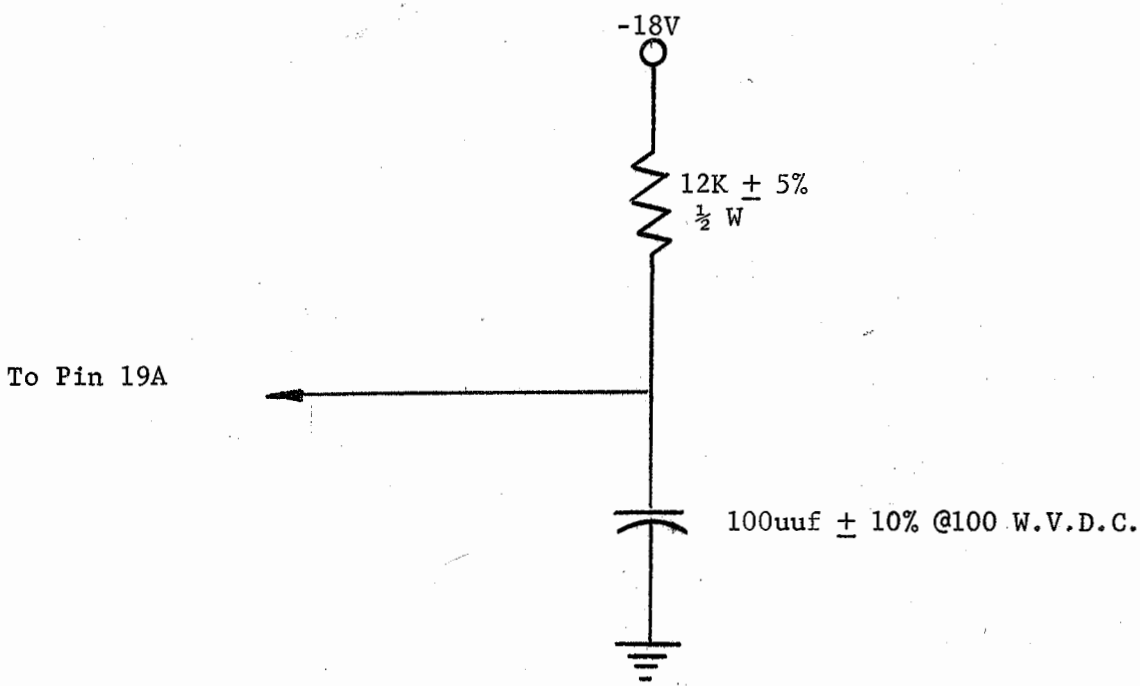


FIGURE 11 LOAD

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